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Teo cauchy goursat convexo

Teorema 1 (de Cauchy-Goursat para convexos).
Sea $\Omega \subset \mathbb{C}$ *un conjunto abierto y convexo y* $f \in \mathcal{H}(\Omega)$*, entonces*

(1) *Existencia de primitiva:* $\exists F \in \mathcal{H}(\Omega)$ *tal que* $F' = f$ *en* Ω*.*

(2) *Teorema de Cauchy-Goursat:* $\forall \gamma$ *camino cerrado en* $\Omega : \displaystyle\int_\gamma f(z)\,\mathrm{d}z = 0$*.*

Observación 1. Para la existencia de primitiva podemos rebajar la hipótesis sobre f y que se siga garantizando la existencia de dicha primitiva: f continua en Ω y $\int_{\partial R} f = 0$ para todo rectángulo $R \subset \Omega \implies \exists F \in \mathcal{H}(\Omega)$ tal que $F' = f$ en Ω.

Referenciado en

- `teo-formula-integral-cauchy-disco`
- `teo-morera-rectangulo`

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